PHYDSC402T — Classical Mechanics

Unit 2: Lagrangian Mechanics

7th Semester, Assam University (FYUG) — Haflong Government College

Paper: PHYDSC402T, Full Marks 100 [ESE 70 + CCA 30] | Unit 2: 12 lectures

Notes 9 of 20 — drafted by gemini-3.6-flash (Kaggle Model Proxy) from the unit syllabus only, without external sources, and checked against the syllabus.

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Unit syllabus (as in the syllabus file)

1. Constraints, generalized coordinates, principle of virtual work, D'Alembert's principle, Lagrange's equations.
2. Velocity dependent potential and dissipation function.
3. First integrals of motion and cyclic coordinates.
4. Hamilton's principle, Lagrange's equations from Hamilton's principle, Hamilton's principle for non-holonomic systems.
5. Symmetry principles and conservation laws.

Top 2 high-mark topics in this unit

1. Hamilton's principle and derivation of Lagrange's equations from Hamilton's principle — the central derivation of the unit, repeated as a long question in Christ College and St. Aloysius papers.
2. Lagrange's equations with velocity-dependent potential and Rayleigh dissipation function, with cyclic coordinates and first integrals of motion — derivation plus application, e.g. charged particle in an electromagnetic field, simple pendulum.

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1. Constraints and Generalised Coordinates

Constraints

In classical mechanics, constraints are defined as physical restrictions or geometrical conditions imposed on the position, velocity, or motion of particles in a system. When a system of particles moves under external influences, its motion may not be completely free; constraints restrict the allowed configurations and trajectories of the system.

Constraints are classified into different types based on their mathematical representation and time dependence.

Holonomic and Non-Holonomic Constraints

A constraint is termed holonomic if the relation expressing the geometrical restriction can be written as an algebraic equation connecting the position coordinates of the particles and time. For a system of N particles with position vectors $\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N$, a holonomic constraint equation has the general form:

$$f(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N, t) = 0$$

If a system has k holonomic constraints, there exist k independent algebraic equations of this form. Holonomic constraints reduce the number of independent coordinates needed to describe the system.

A constraint is termed non-holonomic if it cannot be expressed as an algebraic equation relating coordinates and time. Non-holonomic constraints often appear in the form of non-integrable differential relations involving velocities:

$$\sum_{i=1}^{N} \mathbf{c}_i \cdot d\mathbf{r}_i + c_t dt = 0$$

or as inequalities describing bounded regions of motion:

$$f(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N, t) \geq 0$$

Non-holonomic constraints do not reduce the number of configuration coordinates required to define the state of the system, though they restrict the allowed velocity vectors.

Scleronomic and Rheonomic Constraints

Constraints are also categorized based on whether time explicitly appears in the constraint equation.

A constraint is scleronomic if it is independent of time explicitly. For a scleronomic holonomic constraint, the equation takes the form:

$$f(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N) = 0$$

A constraint is rheonomic if it explicitly depends on time. For a rheonomic holonomic constraint, the equation takes the form:

$$f(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N, t) = 0$$

Examples of Constraints

Consider a simple pendulum consisting of a mass bob suspended from a fixed point by a rigid rod of length l swinging in the xy-plane. Choosing the origin at the point of suspension, the position coordinates (x, y) of the mass bob must satisfy the relation:

$$x^2 + y^2 - l^2 = 0$$

Because this equation connects position coordinates without velocity dependence or explicit time dependence, the simple pendulum constraint is holonomic and scleronomic.

Consider a particle constrained to move on the surface of a rigid sphere of fixed radius R centered at the origin. The coordinates (x, y, z) of the particle satisfy the constraint equation:

$$x^2 + y^2 + z^2 - R^2 = 0$$

This is an algebraic relation independent of velocity and explicit time, making it holonomic and scleronomic. However, if the particle is placed on top of the sphere and is free to fall off, the constraint condition becomes an inequality:

$$x^2 + y^2 + z^2 - R^2 \geq 0$$

In this situation, the constraint is non-holonomic because an inequality cannot be integrated into an equality of coordinates.

Consider a rigid sphere or disk of radius R rolling without slipping on a horizontal plane surface. The condition of no slipping requires that the velocity of the point of contact on the rolling body relative to the surface is identically zero. In differential form, the displacement dx along the path is related to the rotation angle $d\theta$ by:

$$dx - Rd\theta = 0$$

In two-dimensional motion, this differential relation is directly integrable to $x - R\theta = C$, making it holonomic. However, for a three-dimensional disk rolling without slipping on a plane, the rolling conditions relate the spatial displacement components to three orientation angles via differential equations that cannot be integrated into algebraic relations between coordinates. Therefore, rolling without slipping in three dimensions is a non-holonomic constraint.

Generalised Coordinates

Generalised coordinates are defined as any set of independent parameters $q_1, q_2, \ldots, q_n$ that completely and uniquely specify the physical configuration of a system at any time, consistent with all holonomic constraints.

Degrees of Freedom

The number of degrees of freedom of a physical system is defined as the minimum number of independent coordinates required to completely specify its dynamic configuration. For a system composed of N particles moving in three-dimensional space, the total number of initial Cartesian coordinates is $3N$. If the system is subjected to k independent holonomic constraint equations:

$$f_j(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N, t) = 0 \quad (j = 1, 2, \ldots, k)$$

each constraint equation eliminates one coordinate, reducing the required number of independent coordinates. The number of degrees of freedom n is given by:

$$n = 3N - k$$

Transformation from Cartesian Coordinates

For a system of N particles with n degrees of freedom, the position vector $\mathbf{r}_i$ of the i-th particle in standard Cartesian coordinates can be written as a function of the n generalized coordinates $q_1, q_2, \ldots, q_n$ and time t. The transformation equation for the position vector of the i-th particle is:

$$\mathbf{r}_i = \mathbf{r}_i(q_1, q_2, \ldots, q_n, t) \quad (i = 1, 2, \ldots, N)$$

To determine the infinitesimal physical displacement $d\mathbf{r}_i$ of particle i resulting from small changes in the generalized coordinates dq_j over an infinitesimal time interval dt, we apply the chain rule of partial differentiation:

$$d\mathbf{r}_i = \sum_{j=1}^{n} \frac{\partial \mathbf{r}_i}{\partial q_j} dq_j + \frac{\partial \mathbf{r}_i}{\partial t} dt$$

Dividing the infinitesimal displacement vector by dt gives the velocity $\mathbf{v}_i = \dot{\mathbf{r}}_i$ of the i-th particle expressed in terms of generalized coordinates and generalized velocities $\dot{q}_j$:

$$\mathbf{v}_i = \dot{\mathbf{r}}_i = \sum_{j=1}^{n} \frac{\partial \mathbf{r}_i}{\partial q_j} \dot{q}_j + \frac{\partial \mathbf{r}_i}{\partial t}$$

Generalised Coordinates for a System of N Particles with Constraints

For a system of N particles subject to k holonomic constraints, the transformation equations implicitly satisfy all k constraint equations simultaneously. Consequently, the n generalized coordinates $q_1, q_2, \ldots, q_n$ are completely independent of one another. Any arbitrary variation of a single generalized coordinate q_j represents a physically allowable configuration change that does not violate any constraint.

Examples of Generalised Coordinates

For a simple pendulum restricted to move in a vertical plane, the system consists of $N = 1$ particle moving in 2 dimensions with $k = 1$ holonomic constraint ($x^2 + y^2 = l^2$). The number of degrees of freedom is:

$$n = 2(1) - 1 = 1$$

The single generalized coordinate chosen is the angular displacement $q_1 = \theta$ from the vertical. The transformation equations relating Cartesian coordinates to the generalized coordinate are:

$$x = l \sin\theta$$

$$y = l \cos\theta$$

For a particle of mass m constrained to move on a fixed plane surface ($z = 0$), $N = 1$ in 3 dimensions with $k = 1$ constraint ($z = 0$). The degrees of freedom are:

$$n = 3(1) - 1 = 2$$

The configuration can be described using polar coordinates $q_1 = r$ and $q_2 = \theta$ as generalized coordinates. The transformation equations are:

$$x = r\cos\theta$$

$$y = r\sin\theta$$

$$z = 0$$

2. Principle of Virtual Work and D'Alembert's Principle

Principle of Virtual Work

Virtual Displacement

A virtual displacement $\delta\mathbf{r}_i$ of the i-th particle in a system is defined as an imaginary, infinitesimal change in the position vector of the particle consistent with the forces and constraints imposed on the system at a given fixed instant of time t.

Unlike an actual displacement $d\mathbf{r}_i$, which occurs during an elapsed time interval dt during which constraints may change, a virtual displacement takes place with time held frozen ($\delta t = 0$).

Using the transformation equations $\mathbf{r}_i = \mathbf{r}_i(q_1, q_2, \ldots, q_n, t)$, the real differential displacement $d\mathbf{r}_i$ includes a time differential term:

$$d\mathbf{r}_i = \sum_{j=1}^{n} \frac{\partial \mathbf{r}_i}{\partial q_j} dq_j + \frac{\partial \mathbf{r}_i}{\partial t} dt$$

Setting the variation in time to zero ($\delta t = 0$), the virtual displacement $\delta\mathbf{r}_i$ is expressed purely in terms of the independent variations δq_j of the generalized coordinates:

$$\delta\mathbf{r}_i = \sum_{j=1}^{n} \frac{\partial \mathbf{r}_i}{\partial q_j} \delta q_j$$

Virtual Work

The virtual work δW done on a system of N particles during a virtual displacement is defined as the dot product sum of the forces acting on each particle and their respective virtual displacements:

$$\delta W = \sum_{i=1}^{N} \mathbf{F}_i \cdot \delta\mathbf{r}_i$$

Explanation for Ideal Constraints

The total force $\mathbf{F}_i$ acting on the i-th particle can be resolved into the vector sum of the applied force $\mathbf{F}_i^{(a)}$ (such as gravity or external fields) and the constraint force $\mathbf{f}_i$ (such as normal reactions or tension):

$$\mathbf{F}_i = \mathbf{F}_i^{(a)} + \mathbf{f}_i$$

Substituting this decomposition into the expression for total virtual work yields:

$$\delta W = \sum_{i=1}^{N} (\mathbf{F}_i^{(a)} + \mathbf{f}_i) \cdot \delta \mathbf{r}_i$$

Distributing the dot product inside the sum gives:

$$\delta W = \sum_{i=1}^{N} \mathbf{F}_i^{(a)} \cdot \delta \mathbf{r}_i + \sum_{i=1}^{N} \mathbf{f}_i \cdot \delta \mathbf{r}_i$$

Constraints are defined as ideal constraints if the net virtual work done by the forces of constraint during any virtual displacement consistent with the constraints is zero. For ideal constraints, we have the condition:

$$\sum_{i=1}^{N} \mathbf{f}_i \cdot \delta \mathbf{r}_i = 0$$

Examples of ideal constraints include rigid body interatomic forces, smooth sliding contact surfaces without friction (where $\mathbf{f}_i$ is perpendicular to $\delta \mathbf{r}_i$), and rolling without slipping (where the contact point undergoes zero velocity).

Statement of the Principle of Virtual Work

The Principle of Virtual Work states that for a mechanical system with ideal constraints in static equilibrium, the total virtual work done by the applied forces during any arbitrary virtual displacement consistent with the constraints is identically zero.

For a system in equilibrium, the total force on each particle is zero:

$$\mathbf{F}_i = \mathbf{0} \quad (i = 1, 2, \ldots, N)$$

Taking the dot product of this equilibrium condition with $\delta \mathbf{r}_i$ and summing over all particles gives:

$$\sum_{i=1}^{N} \mathbf{F}_i \cdot \delta \mathbf{r}_i = 0$$

Replacing $\mathbf{F}_i$ with the sum of applied forces and constraint forces gives:

$$\sum_{i=1}^{N} \mathbf{F}_i^{(a)} \cdot \delta \mathbf{r}_i + \sum_{i=1}^{N} \mathbf{f}_i \cdot \delta \mathbf{r}_i = 0$$

Applying the ideal constraint condition $\sum_{i=1}^{N} \mathbf{f}_i \cdot \delta \mathbf{r}_i = 0$ reduces this expression directly to the mathematical statement of the Principle of Virtual Work:

$$\delta W^{(a)} = \sum_{i=1}^{N} \mathbf{F}_i^{(a)} \cdot \delta \mathbf{r}_i = 0$$

D'Alembert's Principle

Statement of D'Alembert's Principle

Newton's second law of motion for the i-th particle of mass m_i in a dynamic system is written as:

$$\mathbf{F}_i = \dot{\mathbf{p}}_i$$

where $\mathbf{p}_i = m_i \mathbf{v}_i$ is the linear momentum of the particle, and $\dot{\mathbf{p}}_i = m_i \mathbf{a}_i$ is the time derivative of momentum.

Rearranging this equation by placing all terms on one side gives:

$$\mathbf{F}_i - \dot{\mathbf{p}}_i = \mathbf{0}$$

The quantity $-\dot{\mathbf{p}}_i = -m_i \mathbf{a}_i$ is termed the reversed effective force or inertial force. D'Alembert's principle states that any dynamic system of particles can be transformed into a system in virtual static equilibrium by adding the reversed effective force $-\dot{\mathbf{p}}_i$ to the total force $\mathbf{F}_i$ acting on each particle i.

Extending the Principle of Virtual Work to Dynamics

D'Alembert extended the principle of virtual work from static equilibrium to dynamic systems by considering virtual displacements from dynamic equilibrium configurations.

Taking the dot product of the dynamic equilibrium equation $(\mathbf{F}_i - \dot{\mathbf{p}}_i) = \mathbf{0}$ with an arbitrary virtual displacement $\delta\mathbf{r}_i$ and summing over all N particles gives:

$$\sum_{i=1}^{N} (\mathbf{F}_i - \dot{\mathbf{p}}_i) \cdot \delta\mathbf{r}_i = 0$$

The total force $\mathbf{F}_i$ is resolved into the applied force $\mathbf{F}_i^{(a)}$ and the force of constraint $\mathbf{f}_i$:

$$\mathbf{F}_i = \mathbf{F}_i^{(a)} + \mathbf{f}_i$$

Substituting this force decomposition into the virtual work equation gives:

$$\sum_{i=1}^{N} (\mathbf{F}_i^{(a)} + \mathbf{f}_i - \dot{\mathbf{p}}_i) \cdot \delta\mathbf{r}_i = 0$$

Expanding this summation yields two separate terms:

$$\sum_{i=1}^{N} (\mathbf{F}_i^{(a)} - \dot{\mathbf{p}}_i) \cdot \delta\mathbf{r}_i + \sum_{i=1}^{N} \mathbf{f}_i \cdot \delta\mathbf{r}_i = 0$$

Restricting the discussion to systems with ideal constraints, the work done by constraint forces under virtual displacement vanishes:

$$\sum_{i=1}^{N} \mathbf{f}_i \cdot \delta\mathbf{r}_i = 0$$

Substituting this constraint relation eliminates the unknown constraint forces $\mathbf{f}_i$ completely, yielding D'Alembert's principle in standard form:

$$\sum_{i=1}^{N}(\mathbf{F}_i^{(a)} - \dot{\mathbf{p}}_i) \cdot \delta\mathbf{r}_i = 0$$

D'Alembert's principle provides the foundation of analytical mechanics because it reformulates dynamic problems purely in terms of applied forces $\mathbf{F}_i^{(a)}$ and accelerations, removing the unknown constraint forces from the equation of motion.

3. Lagrange's Equations

D'Alembert's principle provides a formulation of classical mechanics that eliminates the forces of constraint for holonomic systems. For a system of N particles subjected to virtual displacements $\delta\mathbf{r}_i$ consistent with the constraints, D'Alembert's principle is expressed as:

$$\sum_{i=1}^{N}\left(\mathbf{F}_i^{(a)} - \dot{\mathbf{p}}_i\right) \cdot \delta\mathbf{r}_i = 0$$

where $\mathbf{F}_i^{(a)}$ is the net applied force acting on the i-th particle and $\dot{\mathbf{p}}_i = m_i\ddot{\mathbf{r}}_i$ is the time rate of change of linear momentum of the i-th particle.

To express D'Alembert's principle in terms of n independent generalized coordinates $q_1, q_2, \ldots, q_n$, we consider the transformation equations relating position vectors to generalized coordinates and time:

$$\mathbf{r}_i = \mathbf{r}_i(q_1, q_2, \ldots, q_n, t)$$

The virtual displacement $\delta\mathbf{r}_i$ of the i-th particle, which occurs at a fixed instant of time, is given by the differential relation:

$$\delta\mathbf{r}_i = \sum_{j=1}^{n}\frac{\partial\mathbf{r}_i}{\partial q_j}\delta q_j$$

Substituting the virtual displacement into the applied force contribution of D'Alembert's principle gives:

$$\sum_{i=1}^{N}\mathbf{F}_i^{(a)} \cdot \delta\mathbf{r}_i = \sum_{i=1}^{N}\mathbf{F}_i^{(a)} \cdot \sum_{j=1}^{n}\frac{\partial\mathbf{r}_i}{\partial q_j}\delta q_j$$

Interchanging the order of summation yields:

$$\sum_{i=1}^{N}\mathbf{F}_i^{(a)} \cdot \delta\mathbf{r}_i = \sum_{j=1}^{n}\left(\sum_{i=1}^{N}\mathbf{F}_i^{(a)} \cdot \frac{\partial\mathbf{r}_i}{\partial q_j}\right)\delta q_j$$

We define the generalized force Q_j corresponding to the generalized coordinate q_j as:

$$Q_j = \sum_{i=1}^{N} \mathbf{F}_i^{(a)} \cdot \frac{\partial \mathbf{r}_i}{\partial q_j}$$

Thus, the work done by applied forces under a virtual displacement is written as:

$$\sum_{i=1}^{N} \mathbf{F}_i^{(a)} \cdot \delta \mathbf{r}_i = \sum_{j=1}^{n} Q_j \delta q_j$$

Next, we evaluate the inertial term in D'Alembert's principle:

$$\sum_{i=1}^{N} \dot{\mathbf{p}}_i \cdot \delta \mathbf{r}_i = \sum_{i=1}^{N} m_i \ddot{\mathbf{r}}_i \cdot \sum_{j=1}^{n} \frac{\partial \mathbf{r}_i}{\partial q_j} \delta q_j = \sum_{j=1}^{n} \left(\sum_{i=1}^{N} m_i \ddot{\mathbf{r}}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} \right) \delta q_j$$

Applying the product rule of differentiation to the term $m_i \ddot{\mathbf{r}}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j}$, we obtain:

$$\sum_{i=1}^{N} m_i \ddot{\mathbf{r}}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} = \sum_{i=1}^{N} \left[\frac{d}{dt} \left(m_i \dot{\mathbf{r}}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} \right) - m_i \dot{\mathbf{r}}_i \cdot \frac{d}{dt} \left(\frac{\partial \mathbf{r}_i}{\partial q_j} \right) \right]$$

The velocity $\mathbf{v}_i = \dot{\mathbf{r}}_i$ of the i-th particle is given by the total time derivative of $\mathbf{r}_i$:

$$\mathbf{v}_i = \sum_{j=1}^{n} \frac{\partial \mathbf{r}_i}{\partial q_j} \dot{q}_j + \frac{\partial \mathbf{r}_i}{\partial t}$$

Differentiating $\mathbf{v}_i$ partial with respect to the generalized velocity $\dot{q}_j$ yields the identity:

$$\frac{\partial \mathbf{v}_i}{\partial \dot{q}_j} = \frac{\partial \mathbf{r}_i}{\partial q_j}$$

Furthermore, taking the total time derivative of $\frac{\partial \mathbf{r}_i}{\partial q_j}$ gives:

$$\frac{d}{dt} \left(\frac{\partial \mathbf{r}_i}{\partial q_j} \right) = \sum_{k=1}^{n} \frac{\partial^2 \mathbf{r}_i}{\partial q_k \partial q_j} \dot{q}_k + \frac{\partial^2 \mathbf{r}_i}{\partial t \partial q_j}$$

By commuting the order of partial differentiation, this expression is equivalent to:

$$\frac{d}{dt} \left(\frac{\partial \mathbf{r}_i}{\partial q_j} \right) = \frac{\partial \mathbf{v}_i}{\partial q_j}$$

Substituting both kinematic identities into the inertial sum gives:

$$\sum_{i=1}^{N} m_i \ddot{\mathbf{r}}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} = \sum_{i=1}^{N} \left[\frac{d}{dt} \left(m_i \mathbf{v}_i \cdot \frac{\partial \mathbf{v}_i}{\partial \dot{q}_j} \right) - m_i \mathbf{v}_i \cdot \frac{\partial \mathbf{v}_i}{\partial q_j} \right]$$

The kinetic energy T of the system is defined as:

$$T = \frac{1}{2}\sum_{i=1}^{N} m_i v_i^2 = \frac{1}{2}\sum_{i=1}^{N} m_i(\mathbf{v}_i \cdot \mathbf{v}_i)$$

Differentiating T partial with respect to $\dot{q}_j$ produces:

$$\frac{\partial T}{\partial \dot{q}_j} = \sum_{i=1}^{N} m_i \mathbf{v}_i \cdot \frac{\partial \mathbf{v}_i}{\partial \dot{q}_j}$$

Differentiating T partial with respect to q_j produces:

$$\frac{\partial T}{\partial q_j} = \sum_{i=1}^{N} m_i \mathbf{v}_i \cdot \frac{\partial \mathbf{v}_i}{\partial q_j}$$

Thus, the inertial expression simplifies directly to:

$$\sum_{i=1}^{N} m_i \ddot{\mathbf{r}}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} = \frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_j}\right) - \frac{\partial T}{\partial q_j}$$

Substituting the applied and inertial terms back into D'Alembert's principle yields:

$$\sum_{j=1}^{n}\left[\left(\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_j}\right) - \frac{\partial T}{\partial q_j}\right) - Q_j\right]\delta q_j = 0$$

Since the generalized coordinates q_j are independent for a holonomic system, their virtual displacements δq_j are entirely arbitrary and independent. Therefore, the coefficient of each δq_j must vanish independently, giving Lagrange's equations of motion:

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_j}\right) - \frac{\partial T}{\partial q_j} = Q_j, \quad j = 1, 2, \ldots, n$$

For conservative forces, the applied forces are derivable from a scalar potential function $V(\mathbf{r}_1, \ldots, \mathbf{r}_N)$:

$$\mathbf{F}_i^{(a)} = -\nabla_i V$$

The generalized force Q_j then becomes:

$$Q_j = -\sum_{i=1}^{N} \nabla_i V \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} = -\frac{\partial V}{\partial q_j}$$

Substituting this potential representation of generalized force into Lagrange's equations gives:

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_j}\right) - \frac{\partial T}{\partial q_j} = -\frac{\partial V}{\partial q_j}$$

Because the potential energy $V = V(q_1, \ldots, q_n, t)$ is independent of generalized velocities $\dot{q}_j$, we have $\frac{\partial V}{\partial \dot{q}_j} = 0$. Consequently, the derivative $\frac{\partial T}{\partial \dot{q}_j}$ can be written as $\frac{\partial (T-V)}{\partial \dot{q}_j}$. Combining terms yields:

$$\frac{d}{dt}\left(\frac{\partial (T-V)}{\partial \dot{q}_j}\right) - \frac{\partial (T-V)}{\partial q_j} = 0$$

Defining the Lagrangian function L as $L = T - V$, the equations of motion for a conservative system assume the standard form:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right) - \frac{\partial L}{\partial q_j} = 0, \quad j = 1, 2, \ldots, n$$

As an application of Lagrange's equations, consider a simple pendulum consisting of a mass point m suspended by a massless rigid rod of length l under constant gravitational acceleration g. The system has one degree of freedom, described by the angular displacement θ from the downward vertical.

The position coordinates of the mass point in Cartesian space are:

$$x = l\sin\theta$$

$$y = -l\cos\theta$$

Differentiating these coordinates with respect to time gives the velocity components:

$$\dot{x} = l\dot{\theta}\cos\theta$$

$$\dot{y} = l\dot{\theta}\sin\theta$$

The kinetic energy T of the pendulum is calculated as:

$$T = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) = \frac{1}{2}m\left(l^2\dot{\theta}^2\cos^2\theta + l^2\dot{\theta}^2\sin^2\theta\right) = \frac{1}{2}ml^2\dot{\theta}^2$$

Taking the lowest point of the swing $(\theta = 0)$ as the reference level, the potential energy V is:

$$V = mgl(1 - \cos\theta)$$

The Lagrangian $L = T - V$ of the simple pendulum is therefore:

$$L = \frac{1}{2}ml^2\dot{\theta}^2 - mgl(1 - \cos\theta)$$

To apply Lagrange's equation for the generalized coordinate θ, we compute the required partial derivatives:

$$\frac{\partial L}{\partial \dot{\theta}} = ml^2\dot{\theta}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = ml^2\ddot{\theta}$$

$$\frac{\partial L}{\partial \theta} = -mgl\sin\theta$$

Substituting these derivatives into Lagrange's equation $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = 0$ yields:

$$ml^2\ddot{\theta} - (-mgl\sin\theta) = 0$$

$$ml^2\ddot{\theta} + mgl\sin\theta = 0$$

Dividing through by the non-zero quantity ml^2 gives the equation of motion:

$$\ddot{\theta} + \frac{g}{l}\sin\theta = 0$$

For small angular oscillations ($\theta \ll 1$), we use the Taylor expansion approximation $\sin\theta \approx \theta$, which simplifies the equation of motion to:

$$\ddot{\theta} + \frac{g}{l}\theta = 0$$

This is the standard differential equation for a simple harmonic oscillator of angular frequency ω:

$$\omega^2 = \frac{g}{l}$$

$$\omega = \sqrt{\frac{g}{l}}$$

The corresponding period of oscillation T_{period} is given by:

$$T_{\text{period}} = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{l}{g}}$$

4. Velocity-Dependent Potential and Dissipation Function

When the generalized forces Q_j depend on velocities in addition to coordinates, Lagrange's equations can still be expressed in standard form if Q_j can be obtained from a generalized velocity-dependent potential function $U(q_1, \ldots, q_n, \dot{q}_1, \ldots, \dot{q}_n, t)$ defined by the relation:

$$Q_j = -\frac{\partial U}{\partial q_j} + \frac{d}{dt}\left(\frac{\partial U}{\partial \dot{q}_j}\right)$$

Substituting this form of generalized force into Lagrange's equations $\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_j}\right) - \frac{\partial T}{\partial q_j} = Q_j$ gives:

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_j}\right) - \frac{\partial T}{\partial q_j} = -\frac{\partial U}{\partial q_j} + \frac{d}{dt}\left(\frac{\partial U}{\partial \dot{q}_j}\right)$$

Rearranging terms leads to:

$$\frac{d}{dt}\left(\frac{\partial (T-U)}{\partial \dot{q}_j}\right) - \frac{\partial (T-U)}{\partial q_j} = 0$$

Defining the extended Lagrangian function as $L = T - U$, the equations of motion retain the canonical form:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right) - \frac{\partial L}{\partial q_j} = 0$$

A fundamental physical application of a velocity-dependent potential occurs for a particle of mass m and charge q moving in an electromagnetic field described by a scalar potential $\phi(\mathbf{r}, t)$ and a vector potential $\mathbf{A}(\mathbf{r}, t)$. The electromagnetic Lorentz force $\mathbf{F}$ acting on the particle is:

$$\mathbf{F} = q\left(\mathbf{E} + \mathbf{v} \times \mathbf{B}\right)$$

where the electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$ are expressed in terms of the potentials as:

$$\mathbf{E} = -\nabla\phi - \frac{\partial \mathbf{A}}{\partial t}$$

$$\mathbf{B} = \nabla \times \mathbf{A}$$

The generalized velocity-dependent potential U for this interaction is:

$$U = q\phi - q\mathbf{v} \cdot \mathbf{A}$$

The corresponding Lagrangian $L = T - U$ for the charged particle is:

$$L = \frac{1}{2}mv^2 - q\phi + q\mathbf{v} \cdot \mathbf{A}$$

In Cartesian coordinates, the Lagrangian takes the explicit form:

$$L = \frac{1}{2}m\left(\dot{x}^2 + \dot{y}^2 + \dot{z}^2\right) - q\phi(x, y, z, t) + q\left(\dot{x}A_x + \dot{y}A_y + \dot{z}A_z\right)$$

We derive the equation of motion along the x-direction. First, computing the partial derivative of L with respect to $\dot{x}$ gives:

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x} + qA_x$$

Taking the total time derivative of this expression yields:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = m\ddot{x} + q\frac{dA_x}{dt}$$

Expanding the total time derivative of $A_x(x, y, z, t)$ using the chain rule gives:

$$\frac{dA_x}{dt} = \frac{\partial A_x}{\partial t} + \dot{x}\frac{\partial A_x}{\partial x} + \dot{y}\frac{\partial A_x}{\partial y} + \dot{z}\frac{\partial A_x}{\partial z}$$

So the time derivative term becomes:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = m\ddot{x} + q\left(\frac{\partial A_x}{\partial t} + \dot{x}\frac{\partial A_x}{\partial x} + \dot{y}\frac{\partial A_x}{\partial y} + \dot{z}\frac{\partial A_x}{\partial z}\right)$$

Next, computing the partial derivative of L with respect to x gives:

$$\frac{\partial L}{\partial x} = -q\frac{\partial \phi}{\partial x} + q\left(\dot{x}\frac{\partial A_x}{\partial x} + \dot{y}\frac{\partial A_y}{\partial x} + \dot{z}\frac{\partial A_z}{\partial x}\right)$$

Substituting both derivatives into Lagrange's equation $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} = 0$ yields:

$$m\ddot{x} + q\left(\frac{\partial A_x}{\partial t} + \dot{x}\frac{\partial A_x}{\partial x} + \dot{y}\frac{\partial A_x}{\partial y} + \dot{z}\frac{\partial A_x}{\partial z}\right) - \left[-q\frac{\partial \phi}{\partial x} + q\left(\dot{x}\frac{\partial A_x}{\partial x} + \dot{y}\frac{\partial A_y}{\partial x} + \dot{z}\frac{\partial A_z}{\partial x}\right)\right] = 0$$

Canceling the common term $q\dot{x}\frac{\partial A_x}{\partial x}$ and rearranging terms isolates $m\ddot{x}$:

$$m\ddot{x} = q\left(-\frac{\partial \phi}{\partial x} - \frac{\partial A_x}{\partial t}\right) + q\dot{y}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right) + q\dot{z}\left(\frac{\partial A_z}{\partial x} - \frac{\partial A_x}{\partial z}\right)$$

Recalling the definitions of the electric field component E_x and magnetic field components B_y, B_z:

$$E_x = -\frac{\partial \phi}{\partial x} - \frac{\partial A_x}{\partial t}$$

$$B_z = \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}$$

$$B_y = \frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}$$

Substituting these field definitions back into the force expression yields:

$$m\ddot{x} = qE_x + q\left(\dot{y}B_z - \dot{z}B_y\right)$$

Recognizing $\left(\dot{y}B_z - \dot{z}B_y\right)$ as the x-component of the vector cross product $\mathbf{v} \times \mathbf{B}$, we get:

$$m\ddot{x} = qE_x + q(\mathbf{v} \times \mathbf{B})_x$$

Repeating this calculation symmetrically for y and z coordinates gives the vector equation of motion:

$$m\frac{d\mathbf{v}}{dt} = q\left(\mathbf{E} + \mathbf{v} \times \mathbf{B}\right)$$

This demonstrates that the velocity-dependent potential $U = q\phi - q\mathbf{v} \cdot \mathbf{A}$ correctly yields the Lorentz force equation of motion.

For systems involving frictional forces proportional to velocity, Lord Rayleigh introduced a dissipation function F. For a system described by generalized velocities $\dot{q}_j$, the Rayleigh dissipation function F is defined as a quadratic form:

$$F = \frac{1}{2}\sum_{j=1}^{n}\sum_{k=1}^{n} k_{jk}\dot{q}_j\dot{q}_k$$

where $k_{jk} = k_{kj}$ are real symmetric coefficients forming a positive-definite matrix. The non-conservative generalized dissipative force $Q_{j,\text{dis}}$ is derived from F by differentiation:

$$Q_{j,\text{dis}} = -\frac{\partial F}{\partial \dot{q}_j}$$

Including both conservative forces (via $L = T - V$) and dissipative forces in Lagrange's equations gives:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right) - \frac{\partial L}{\partial q_j} = Q_{j,\text{dis}} = -\frac{\partial F}{\partial \dot{q}_j}$$

Rearranging terms gives Lagrange's equations with dissipation:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right) - \frac{\partial L}{\partial q_j} + \frac{\partial F}{\partial \dot{q}_j} = 0, \quad j = 1, 2, \ldots, n$$

As an application of the Rayleigh dissipation function, consider a one-dimensional damped harmonic oscillator consisting of a particle of mass m, attached to a spring of stiffness k, moving in a viscous medium with damping constant b.

The kinetic energy T, potential energy V, and Rayleigh dissipation function F are:

$$T = \frac{1}{2}m\dot{x}^2$$

$$V = \frac{1}{2}kx^2$$

$$F = \frac{1}{2}b\dot{x}^2$$

The Lagrangian is:

$$L = T - V = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2$$

We compute the required partial derivatives for the equation of motion:

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) = m\ddot{x}$$

$$\frac{\partial L}{\partial x} = -kx$$

$$\frac{\partial F}{\partial \dot{x}} = b\dot{x}$$

Substituting these expressions into Lagrange's equation with dissipation $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x} + \frac{\partial F}{\partial \dot{x}} = 0$ yields:

$$m\ddot{x} - (-kx) + b\dot{x} = 0$$

$$m\ddot{x} + b\dot{x} + kx = 0$$

This recovers the standard differential equation governing a linearly damped harmonic oscillator.

5. First Integrals of Motion and Cyclic Coordinates

In Lagrangian mechanics, a generalized coordinate q_k is defined as a cyclic (or ignorable) coordinate if it does not explicitly appear in the Lagrangian function $L(q_1, \ldots, q_n, \dot{q}_1, \ldots, \dot{q}_n, t)$, even though its time derivative $\dot{q}_k$ may be present. Mathematically, q_k is cyclic if:

$$\frac{\partial L}{\partial q_k} = 0$$

For any generalized coordinate q_j, the corresponding generalized momentum p_j conjugate to q_j is defined as:

$$p_j = \frac{\partial L}{\partial \dot{q}_j}$$

Lagrange's equation of motion for coordinate q_j can be rewritten using the generalized momentum p_j:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right) - \frac{\partial L}{\partial q_j} = 0$$

$$\frac{dp_j}{dt} = \frac{\partial L}{\partial q_j}$$

If the generalized coordinate q_j is cyclic, $\frac{\partial L}{\partial q_j} = 0$, which reduces the momentum equation to:

$$\frac{dp_j}{dt} = 0$$

Integrating this equation with respect to time gives:

$$p_j = \text{constant}$$

Thus, the generalized momentum conjugate to a cyclic coordinate is a constant of motion, providing a first integral of motion.

Next, we derive the energy first integral for systems where the Lagrangian L has no explicit time dependence. Consider a general Lagrangian $L(q_1, \ldots, q_n, \dot{q}_1, \ldots, \dot{q}_n, t)$. The total time derivative of L is given by the chain rule:

$$\frac{dL}{dt} = \sum_{j=1}^{n} \frac{\partial L}{\partial q_j}\dot{q}_j + \sum_{j=1}^{n} \frac{\partial L}{\partial \dot{q}_j}\ddot{q}_j + \frac{\partial L}{\partial t}$$

Using Lagrange's equations, we substitute $\frac{\partial L}{\partial q_j} = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right)$ into the first sum:

$$\frac{dL}{dt} = \sum_{j=1}^{n} \left[\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right)\dot{q}_j + \frac{\partial L}{\partial \dot{q}_j}\ddot{q}_j\right] + \frac{\partial L}{\partial t}$$

Applying the product rule of differentiation to the term inside the bracket:

$$\frac{d}{dt}\left(\dot{q}_j\frac{\partial L}{\partial \dot{q}_j}\right) = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right)\dot{q}_j + \frac{\partial L}{\partial \dot{q}_j}\ddot{q}_j$$

We can rewrite the total time derivative of L as:

$$\frac{dL}{dt} = \sum_{j=1}^{n}\frac{d}{dt}\left(\dot{q}_j\frac{\partial L}{\partial \dot{q}_j}\right) + \frac{\partial L}{\partial t}$$

Rearranging terms to consolidate the total time derivatives yields:

$$\frac{d}{dt}\left(\sum_{j=1}^{n}\dot{q}_j\frac{\partial L}{\partial \dot{q}_j} - L\right) = -\frac{\partial L}{\partial t}$$

We define the energy integral function h (also known as Jacobi's integral) as:

$$h(q,\dot{q}) = \sum_{j=1}^{n}\dot{q}_j\frac{\partial L}{\partial \dot{q}_j} - L$$

The rate of change of h with respect to time is:

$$\frac{dh}{dt} = -\frac{\partial L}{\partial t}$$

If the Lagrangian L does not depend explicitly on time t, then $\frac{\partial L}{\partial t} = 0$, which yields:

$$\frac{dh}{dt} = 0$$

$$h = \text{constant}$$

Therefore, when the Lagrangian has no explicit dependence on time, h is a first integral of motion.

To connect h to the total mechanical energy $E = T + V$, consider a system where the transformation equations to Cartesian coordinates do not depend explicitly on time ($\frac{\partial \mathbf{r}_i}{\partial t} = 0$). In this case, kinetic energy T is a homogeneous quadratic function of the generalized velocities $\dot{q}_j$:

$$T = \frac{1}{2}\sum_{j=1}^{n}\sum_{k=1}^{n}a_{jk}(q)\dot{q}_j\dot{q}_k$$

By Euler's theorem for homogeneous functions of degree 2, T satisfies:

$$\sum_{j=1}^{n}\dot{q}_j\frac{\partial T}{\partial \dot{q}_j} = 2T$$

For a conservative potential $V(q)$ independent of velocities, $\frac{\partial L}{\partial \dot{q}_j} = \frac{\partial T}{\partial \dot{q}_j}$. Substituting this identity into the expression for h:

$$h = \sum_{j=1}^{n} \dot{q}_j \frac{\partial T}{\partial \dot{q}_j} - L$$

$$h = 2T - (T - V)$$

$$h = T + V = E$$

Thus, under natural conservative constraints without explicit time dependence, the conserved quantity h represents the constant total mechanical energy E of the system.

6. Hamilton's Principle

The formulation of classical mechanics can be based on integral principles in addition to differential equations of motion. The fundamental integral principle governing the motion of a mechanical system is Hamilton's principle, which expresses the equations of motion in terms of a single scalar functional known as the action integral.

Consider a dynamical system with n degrees of freedom described by a set of n independent generalized coordinates q_j ($j = 1, 2, \ldots, n$). The instantaneous state of the system is represented by a single point in an n-dimensional configuration space. As time progresses from an initial time t_1 to a final time t_2, the system traces out a continuous path in configuration space.

To analyze the path taken by the system, we define a line integral called the action integral, denoted by S. The action integral is defined as the time integral of the Lagrangian function $L(q_j, \dot{q}_j, t)$ evaluated between the initial time t_1 and the final time t_2:

$$S = \int_{t_1}^{t_2} L(q_1, q_2, \ldots, q_n, \dot{q}_1, \dot{q}_2, \ldots, \dot{q}_n, t)\, dt$$

The core idea of the calculus of variations is to compare the action integral along the actual physical path traversed by the system with the action integral along neighboring, geometrically possible paths. Let $q_j(t)$ denote the coordinates defining the actual physical path. A varied path is constructed by adding a small variation to the true path:

$$q_j(t, \alpha) = q_j(t, 0) + \alpha \eta_j(t)$$

Here, $q_j(t, 0) = q_j(t)$ represents the actual path, α is a small continuous parameter, and $\eta_j(t)$ are arbitrary continuously differentiable functions of time representing the variations.

A fundamental requirement of Hamilton's principle is the end-point condition. The varied paths are chosen such that they pass through the exact same initial configuration at time t_1 and the exact same final configuration at time t_2 as the actual physical path. Therefore, the variation functions $\eta_j(t)$ must vanish at the boundary times t_1 and t_2:

$$\eta_j(t_1) = 0 \quad \text{for all } j = 1, 2, \ldots, n$$

$$\eta_j(t_2) = 0 \quad \text{for all } j = 1, 2, \ldots, n$$

In terms of the virtual variation operator δ, defined as $\delta q_j = \alpha \eta_j(t)$, the end-point conditions state that the variations of the generalized coordinates vanish at the end points:

$$\delta q_j(t_1) = 0 \quad \text{for all } j = 1, 2, \ldots, n$$

$$\delta q_j(t_2) = 0 \quad \text{for all } j = 1, 2, \ldots, n$$

Hamilton's principle states that of all possible paths between two fixed configurations at times t_1 and t_2, the actual path traversed by a holonomic, monogenic system is the one for which the action integral S is stationary. In variational notation, the first variation of the action integral along the actual path is equal to zero:

$$\delta S = \delta \int_{t_1}^{t_2} L(q_j, \dot{q}_j, t)\, dt = 0$$

This stationarity condition means that the action integral has an extremum (typically a minimum) for the trajectory that the system physically follows.

7. Lagrange's Equations from Hamilton's Principle

Lagrange's equations of motion for a conservative, holonomic system can be derived directly from Hamilton's principle using the techniques of the calculus of variations.

We begin with the mathematical statement of Hamilton's principle for a system with n generalized coordinates:

$$\delta S = \delta \int_{t_1}^{t_2} L(q_j, \dot{q}_j, t)\, dt = 0$$

Since the integration limits t_1 and t_2 are fixed and independent of the variation, the variation operator δ commutes with the time integration operator:

$$\delta S = \int_{t_1}^{t_2} \delta L(q_j, \dot{q}_j, t)\, dt = 0$$

The total variation of the Lagrangian function $L(q_1, \ldots, q_n, \dot{q}_1, \ldots, \dot{q}_n, t)$ due to virtual variations δq_j in the coordinates and $\delta \dot{q}_j$ in the velocities is expanded using the chain rule:

$$\delta L = \sum_{j=1}^{n} \left(\frac{\partial L}{\partial q_j} \delta q_j + \frac{\partial L}{\partial \dot{q}_j} \delta \dot{q}_j \right)$$

Substituting this total variation into the integral for δS gives:

$$\delta S = \int_{t_1}^{t_2} \sum_{j=1}^{n} \left(\frac{\partial L}{\partial q_j} \delta q_j + \frac{\partial L}{\partial \dot{q}_j} \delta \dot{q}_j \right) dt = 0$$

By interchanging the order of summation and integration, the variation of the action takes the form:

$$\delta S = \sum_{j=1}^{n} \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q_j} \delta q_j + \frac{\partial L}{\partial \dot{q}_j} \delta \dot{q}_j \right) dt = 0$$

We rewrite the generalized velocity variation $\delta \dot{q}_j$ as the time derivative of the coordinate variation δq_j:

$$\delta \dot{q}_j = \delta \left(\frac{dq_j}{dt} \right) = \frac{d}{dt}(\delta q_j)$$

Substituting this identity into the expression for δS yields:

$$\delta S = \sum_{j=1}^{n} \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q_j} \delta q_j + \frac{\partial L}{\partial \dot{q}_j} \frac{d}{dt}(\delta q_j) \right) dt = 0$$

We evaluate the term involving $\frac{d}{dt}(\delta q_j)$ in each integral by integrating by parts. Taking $u = \frac{\partial L}{\partial \dot{q}_j}$ and $dv = \frac{d}{dt}(\delta q_j) dt$, we get:

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}_j} \frac{d}{dt}(\delta q_j)\, dt = \left[\frac{\partial L}{\partial \dot{q}_j} \delta q_j \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) \delta q_j \, dt$$

We now apply the end-point conditions of Hamilton's principle, which mandate that $\delta q_j(t_1) = 0$ and $\delta q_j(t_2) = 0$. Consequently, the integrated boundary term vanishes identically:

$$\left[\frac{\partial L}{\partial \dot{q}_j} \delta q_j \right]_{t_1}^{t_2} = \frac{\partial L}{\partial \dot{q}_j}(t_2) \delta q_j(t_2) - \frac{\partial L}{\partial \dot{q}_j}(t_1) \delta q_j(t_1) = 0$$

Substituting this result back into the integration-by-parts expression simplifies it to:

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}_j} \frac{d}{dt}(\delta q_j)\, dt = - \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) \delta q_j \, dt$$

Replacing the velocity variation term in the expression for δS with this integrated expression gives:

$$\delta S = \sum_{j=1}^{n} \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q_j} \delta q_j - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) \delta q_j \right) dt = 0$$

Factoring out the coordinate variation δq_j inside the integral yields:

$$\delta S = \int_{t_1}^{t_2} \sum_{j=1}^{n} \left(\frac{\partial L}{\partial q_j} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) \right) \delta q_j \, dt = 0$$

Multiplying the entire expression by -1 transforms the equation into the standard form:

$$\int_{t_1}^{t_2} \sum_{j=1}^{n} \left(\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} \right) \delta q_j \, dt = 0$$

For a holonomic system, all the generalized coordinates q_j are completely independent of one another. Therefore, the variations $\delta q_j(t)$ are arbitrary independent functions between the endpoints t_1 and t_2. By the fundamental lemma of the calculus of variations, for the integral to vanish for any arbitrary choice of independent variations δq_j, the coefficient of each δq_j in the integrand must vanish individually at every instant of time.

This requirement gives Lagrange's equations of motion for each generalized coordinate $j = 1, 2, \ldots, n$:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right) - \frac{\partial L}{\partial q_j} = 0$$

8. Hamilton's Principle for Non-Holonomic Systems

In the standard formulation of Hamilton's principle, the action integral I is defined as the time integral of the Lagrangian $L(q_j, \dot{q}_j, t)$ between two fixed times t_1 and t_2:

$$I = \int_{t_1}^{t_2} L(q_j, \dot{q}_j, t)\, dt$$

For a system subject to holonomic constraints, the constraint equations can be expressed as algebraic relations among the generalized coordinates and time of the form $f_k(q_1, q_2, \ldots, q_n, t) = 0$. These relations can be used to eliminate the dependent coordinates, leaving a set of independent generalized coordinates. Consequently, the virtual variations δq_j are completely independent of one another. The stationary action condition $\delta I = 0$ then yields the standard Euler-Lagrange equations directly by setting the coefficient of each independent variation δq_j to zero.

However, when the system is subject to non-holonomic constraints, the constraint relations are non-integrable differential expressions connecting the generalized coordinates, their velocities, and time. Such constraints cannot be used to eliminate dependent coordinates algebraically. Non-holonomic constraints are typically expressed in differential form as:

$$\sum_{j=1}^{n} a_{lj}\, dq_j + a_{lt}\, dt = 0 \quad (l = 1, 2, \ldots, m)$$

where m is the number of non-holonomic constraints and a_{lj} as well as a_{lt} are functions of the coordinates q_j and time t. For virtual displacements, the variation occurs at a fixed instant of time, which implies that $\delta t = 0$. Thus, the virtual displacements δq_j must satisfy the restrictive relations:

$$\sum_{j=1}^{n} a_{lj}\, \delta q_j = 0 \quad (l = 1, 2, \ldots, m)$$

Because of these relations, the variations δq_j are no longer independent of one another. As a result, the standard form of Hamilton's principle cannot be applied directly by setting the individual coefficients of δq_j to zero.

To incorporate non-holonomic constraints into Hamilton's principle, the method of Lagrange's undetermined multipliers is employed. We begin by taking the variation of the action integral I:

$$\delta I = \delta \int_{t_1}^{t_2} L(q_1, \ldots, q_n, \dot{q}_1, \ldots, \dot{q}_n, t)\, dt = 0$$

Carrying the variation inside the integral sign yields:

$$\int_{t_1}^{t_2} \sum_{j=1}^{n} \left(\frac{\partial L}{\partial q_j}\delta q_j + \frac{\partial L}{\partial \dot{q}_j}\delta \dot{q}_j \right) dt = 0$$

We rewrite the variation of the velocity using the identity $\delta\dot{q}_j = \frac{d}{dt}(\delta q_j)$:

$$\int_{t_1}^{t_2} \sum_{j=1}^{n} \left(\frac{\partial L}{\partial q_j}\delta q_j + \frac{\partial L}{\partial \dot{q}_j}\frac{d}{dt}(\delta q_j) \right) dt = 0$$

Integrating the second term by parts over time gives:

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}_j}\frac{d}{dt}(\delta q_j)\, dt = \left[\frac{\partial L}{\partial \dot{q}_j}\delta q_j \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j} \right) \delta q_j \, dt$$

Since the virtual variations vanish at the endpoints t_1 and t_2, the boundary term disappears:

$$\left[\frac{\partial L}{\partial \dot{q}_j}\delta q_j \right]_{t_1}^{t_2} = 0$$

Substituting the integrated result back into the variational integral produces:

$$\int_{t_1}^{t_2} \sum_{j=1}^{n} \left[\frac{\partial L}{\partial q_j} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j} \right) \right] \delta q_j \, dt = 0$$

Now, we introduce m undetermined functions of time $\lambda_l(t)$, known as Lagrange multipliers. Each virtual constraint equation is multiplied by its corresponding multiplier $\lambda_l(t)$:

$$\lambda_l(t) \sum_{j=1}^{n} a_{lj}\, \delta q_j = 0$$

Integrating this expression over time from t_1 to t_2 gives:

$$\int_{t_1}^{t_2} \lambda_l(t) \sum_{j=1}^{n} a_{lj}\, \delta q_j \, dt = 0$$

Summing this relation over all m constraints yields:

$$\int_{t_1}^{t_2} \sum_{l=1}^{m} \lambda_l(t) \sum_{j=1}^{n} a_{lj}\, \delta q_j \, dt = 0$$

Exchanging the order of summation gives:

$$\int_{t_1}^{t_2} \sum_{j=1}^{n} \left(\sum_{l=1}^{m} \lambda_l a_{lj} \right) \delta q_j \, dt = 0$$

We now add this constraint integral to the variational equation obtained from Hamilton's principle:

$$\int_{t_1}^{t_2} \sum_{j=1}^{n} \left[\frac{\partial L}{\partial q_j} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j} \right) + \sum_{l=1}^{m} \lambda_l a_{lj} \right] \delta q_j \, dt = 0$$

Although the n variations δq_j are not all independent, we can choose the m multipliers $\lambda_1, \lambda_2, \ldots, \lambda_m$ such that the coefficients of m dependent variations vanish identically inside the integrand. The remaining $n - m$ variations can then be varied independently. Consequently, every term in the bracket must equal zero individually for all $j = 1, 2, \ldots, n$:

$$\frac{\partial L}{\partial q_j} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right) + \sum_{l=1}^{m} \lambda_l a_{lj} = 0$$

Rearranging this equation gives Lagrange's equations for non-holonomic systems with undetermined multipliers:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right) - \frac{\partial L}{\partial q_j} = \sum_{l=1}^{m} \lambda_l a_{lj} \quad (j = 1, 2, \ldots, n)$$

The terms on the right-hand side, $Q_j^c = \sum_{l=1}^{m} \lambda_l a_{lj}$, represent the generalized forces of constraint associated with the non-holonomic constraints acting on the system.

9. Symmetry Principles and Conservation Laws

In analytical mechanics, a symmetry principle refers to the invariance of the Lagrangian of a mechanical system under a continuous spatial or temporal transformation. Every such invariance property gives rise to a corresponding conservation law.

The underlying mechanism linking symmetries to conservation laws can be directly understood using the concept of cyclic coordinates. If a generalized coordinate q_k does not appear explicitly in the Lagrangian L, so that:

$$\frac{\partial L}{\partial q_k} = 0$$

then q_k is defined as a cyclic (or ignorable) coordinate. Substituting this condition into the standard Lagrange equation of motion:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_k}\right) - \frac{\partial L}{\partial q_k} = 0$$

yields:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_k}\right) = 0$$

Defining the generalized momentum conjugate to q_k as:

$$p_k = \frac{\partial L}{\partial \dot{q}_k}$$

we obtain the conservation equation:

$$\frac{dp_k}{dt} = 0 \implies p_k = \text{constant}$$

Thus, the absence of a coordinate from the Lagrangian implies that the conjugate generalized momentum is a conserved quantity.

Homogeneity of Space and Conservation of Linear Momentum

Homogeneity of space means that the properties of an isolated mechanical system remain invariant under an arbitrary displacement of the entire system in space. Consider an arbitrary infinitesimal spatial translation of a system of N particles by a constant vector $\delta\mathbf{r}$. Under this transformation, the position vector of each particle $\mathbf{r}_i$ transforms as:

$$\mathbf{r}_i \to \mathbf{r}_i' = \mathbf{r}_i + \delta\mathbf{r}$$

Taking the time derivative shows that the velocities remain unchanged:

$$\dot{\mathbf{r}}_i' = \dot{\mathbf{r}}_i$$

The change in the position vector of particle i is:

$$\delta\mathbf{r}_i = \delta\mathbf{r}$$

The variation in the Lagrangian $L(\mathbf{r}_i, \dot{\mathbf{r}}_i)$ due to this infinitesimal spatial displacement is given by:

$$\delta L = \sum_{i=1}^{N} \frac{\partial L}{\partial \mathbf{r}_i} \cdot \delta\mathbf{r}_i + \sum_{i=1}^{N} \frac{\partial L}{\partial \dot{\mathbf{r}}_i} \cdot \delta\dot{\mathbf{r}}_i$$

Since the velocities are unaltered, $\delta\dot{\mathbf{r}}_i = 0$, reducing the variation to:

$$\delta L = \sum_{i=1}^{N} \frac{\partial L}{\partial \mathbf{r}_i} \cdot \delta\mathbf{r}$$

Factoring out the constant displacement vector $\delta\mathbf{r}$ yields:

$$\delta L = \left(\sum_{i=1}^{N} \frac{\partial L}{\partial \mathbf{r}_i} \right) \cdot \delta\mathbf{r}$$

By the principle of homogeneity of space, the Lagrangian must be invariant under an arbitrary translation, so $\delta L = 0$. Because $\delta\mathbf{r}$ is arbitrary, we must have:

$$\sum_{i=1}^{N} \frac{\partial L}{\partial \mathbf{r}_i} = 0$$

Applying Lagrange's equation for each particle position vector:

$$\frac{\partial L}{\partial \mathbf{r}_i} = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\mathbf{r}}_i} \right)$$

we can substitute this into the sum:

$$\sum_{i=1}^{N} \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\mathbf{r}}_i}\right) = 0$$

Interchanging the order of summation and time differentiation gives:

$$\frac{d}{dt}\left(\sum_{i=1}^{N} \frac{\partial L}{\partial \dot{\mathbf{r}}_i}\right) = 0$$

Recognizing the conjugate momentum of particle i as $\mathbf{p}_i = \frac{\partial L}{\partial \dot{\mathbf{r}}_i}$, this equation becomes:

$$\frac{d}{dt}\left(\sum_{i=1}^{N} \mathbf{p}_i\right) = 0$$

Defining the total linear momentum of the system as $\mathbf{P} = \sum_{i=1}^{N} \mathbf{p}_i$, we arrive at the conservation law:

$$\frac{d\mathbf{P}}{dt} = 0 \implies \mathbf{P} = \text{constant}$$

From the cyclic-coordinate perspective, if the Lagrangian is independent of translations along a particular direction (say, x), then x is a cyclic coordinate, and the component of total linear momentum P_x along that direction is conserved.

Isotropy of Space and Conservation of Angular Momentum

Isotropy of space means that the properties of an isolated system are identical in all directions, so the Lagrangian is invariant under an arbitrary infinitesimal rotation of the system in space. Let an infinitesimal rotation be defined by a vector $\delta\boldsymbol{\theta}$, whose magnitude represents the angle of rotation and whose direction specifies the axis of rotation.

Under this rotation, the position vector $\mathbf{r}_i$ and velocity vector $\dot{\mathbf{r}}_i$ of particle i change according to:

$$\delta\mathbf{r}_i = \delta\boldsymbol{\theta} \times \mathbf{r}_i$$

$$\delta\dot{\mathbf{r}}_i = \delta\boldsymbol{\theta} \times \dot{\mathbf{r}}_i$$

The variation in the Lagrangian L due to this rotation is expressed as:

$$\delta L = \sum_{i=1}^{N} \left(\frac{\partial L}{\partial \mathbf{r}_i} \cdot \delta\mathbf{r}_i + \frac{\partial L}{\partial \dot{\mathbf{r}}_i} \cdot \delta\dot{\mathbf{r}}_i\right)$$

Substituting the canonical momentum $\mathbf{p}_i = \frac{\partial L}{\partial \dot{\mathbf{r}}_i}$ and Lagrange's equation $\dot{\mathbf{p}}_i = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\mathbf{r}}_i}\right) = \frac{\partial L}{\partial \mathbf{r}_i}$ into the expression for δL yields:

$$\delta L = \sum_{i=1}^{N} \left[\dot{\mathbf{p}}_i \cdot (\delta\boldsymbol{\theta} \times \mathbf{r}_i) + \mathbf{p}_i \cdot (\delta\boldsymbol{\theta} \times \dot{\mathbf{r}}_i)\right]$$

Using the scalar triple product identity $\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A})$, we rewrite the two terms inside the summation:

$$\dot{\mathbf{p}}_i \cdot (\delta\boldsymbol{\theta} \times \mathbf{r}_i) = \delta\boldsymbol{\theta} \cdot (\mathbf{r}_i \times \dot{\mathbf{p}}_i)$$

$$\mathbf{p}_i \cdot (\delta\boldsymbol{\theta} \times \dot{\mathbf{r}}_i) = \delta\boldsymbol{\theta} \cdot (\dot{\mathbf{r}}_i \times \mathbf{p}_i)$$

Substituting these identities back into the variation of L gives:

$$\delta L = \sum_{i=1}^{N} \left[\delta\boldsymbol{\theta} \cdot (\mathbf{r}_i \times \dot{\mathbf{p}}_i) + \delta\boldsymbol{\theta} \cdot (\dot{\mathbf{r}}_i \times \mathbf{p}_i) \right]$$

Factoring out the constant infinitesimal rotation vector $\delta\boldsymbol{\theta}$ yields:

$$\delta L = \delta\boldsymbol{\theta} \cdot \sum_{i=1}^{N} (\mathbf{r}_i \times \dot{\mathbf{p}}_i + \dot{\mathbf{r}}_i \times \mathbf{p}_i)$$

Applying the product rule of differentiation to the term in parentheses:

$$\frac{d}{dt}(\mathbf{r}_i \times \mathbf{p}_i) = \dot{\mathbf{r}}_i \times \mathbf{p}_i + \mathbf{r}_i \times \dot{\mathbf{p}}_i$$

Substituting this derivative into the expression for δL gives:

$$\delta L = \delta\boldsymbol{\theta} \cdot \sum_{i=1}^{N} \frac{d}{dt}(\mathbf{r}_i \times \mathbf{p}_i)$$

Interchanging the summation and time derivative produces:

$$\delta L = \delta\boldsymbol{\theta} \cdot \frac{d}{dt} \left(\sum_{i=1}^{N} \mathbf{r}_i \times \mathbf{p}_i \right)$$

By isotropy of space, $\delta L = 0$ for any arbitrary rotation vector $\delta\boldsymbol{\theta}$, which requires:

$$\frac{d}{dt} \left(\sum_{i=1}^{N} \mathbf{r}_i \times \mathbf{p}_i \right) = 0$$

Defining the total angular momentum of the system as $\mathbf{L}_{tot} = \sum_{i=1}^{N} (\mathbf{r}_i \times \mathbf{p}_i)$, we get:

$$\frac{d\mathbf{L}_{tot}}{dt} = 0 \implies \mathbf{L}_{tot} = \text{constant}$$

In terms of cyclic coordinates, if the system is symmetric under rotation about a specific axis, the angular coordinate θ describing rotation about that axis does not appear in L. Thus, θ is cyclic, and the conjugate momentum $p_\theta = L_{axis}$ (the component of angular momentum along that axis) is conserved.

Homogeneity of Time and Conservation of Energy

Homogeneity of time implies that time is uniform and that the intrinsic laws governing the mechanical system do not depend on the choice of the time origin. Consequently, the Lagrangian $L(q_j, \dot{q}_j)$ does not depend explicitly on time t, meaning its partial derivative with respect to time vanishes:

$$\frac{\partial L}{\partial t} = 0$$

To determine the conserved quantity, we take the total time derivative of $L(q_j, \dot{q}_j, t)$:

$$\frac{dL}{dt} = \sum_{j=1}^{n} \frac{\partial L}{\partial q_j}\dot{q}_j + \sum_{j=1}^{n} \frac{\partial L}{\partial \dot{q}_j}\ddot{q}_j + \frac{\partial L}{\partial t}$$

We substitute Lagrange's equations of motion, $\frac{\partial L}{\partial q_j} = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right)$, into the first summation:

$$\frac{dL}{dt} = \sum_{j=1}^{n} \left[\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right)\right]\dot{q}_j + \sum_{j=1}^{n} \frac{\partial L}{\partial \dot{q}_j}\ddot{q}_j + \frac{\partial L}{\partial t}$$

Combining the first two summation terms using the product rule of differentiation gives:

$$\frac{dL}{dt} = \frac{d}{dt}\left(\sum_{j=1}^{n} \dot{q}_j \frac{\partial L}{\partial \dot{q}_j}\right) + \frac{\partial L}{\partial t}$$

Rearranging this equation to bring all total time derivatives to one side produces:

$$\frac{d}{dt}\left(\sum_{j=1}^{n} \dot{q}_j \frac{\partial L}{\partial \dot{q}_j} - L\right) = -\frac{\partial L}{\partial t}$$

By the homogeneity of time, $\frac{\partial L}{\partial t} = 0$. Therefore, the equation reduces to:

$$\frac{d}{dt}\left(\sum_{j=1}^{n} \dot{q}_j \frac{\partial L}{\partial \dot{q}_j} - L\right) = 0$$

The quantity inside the parentheses is known as the energy function or Jacobi integral h:

$$h = \sum_{j=1}^{n} \dot{q}_j \frac{\partial L}{\partial \dot{q}_j} - L$$

Integrating with respect to time yields the conservation law:

$$h = \text{constant}$$

When the transformation equations connecting Cartesian coordinates to generalized coordinates are independent of time, and the potential energy V is velocity-independent, h equals the total mechanical energy $E = T + V$. Thus, the homogeneity of time implies the conservation of total mechanical energy. In the framework of cyclic coordinates, the explicit absence of time from the Lagrangian ensures that the energy function h remains constant throughout the motion.

Questions

Short form first (2–3 marks), then long form (5–10 marks). **No Assam University questions were found for this unit in the paper bank** — the questions below are from other universities' papers and standard syllabus-derived questions, marked as such.

Short-form questions

S1. [standard] Define constraints; distinguish holonomic and non-holonomic constraints.

Answer in brief: Section 1 — constraints are restrictions on the motion; holonomic constraints can be written as equations $f(q_1, \dots, q_n, t) = 0$ between coordinates, non-holonomic ones cannot (e.g. rolling without slipping).

S2. [standard] State the principle of virtual work.

Answer in brief: Section 2 — for a system in equilibrium the total virtual work of the applied forces vanishes, $\sum \mathbf{F}_i \cdot \delta\mathbf{r}_i = 0$, for any virtual displacement consistent with the constraints.

S3. [standard] State D'Alembert's principle.

Answer in brief: Section 2 — adding the reversed effective force $-m_i\mathbf{a}_i$ to the applied forces puts the system in (dynamic) equilibrium: $\sum(\mathbf{F}_i - m_i\mathbf{a}_i)\cdot\delta\mathbf{r}_i = 0$.

S4. [standard] Write Lagrange's equations for a system with n generalised coordinates.

Answer in brief: Section 3 — $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_j}\right) - \frac{\partial L}{\partial q_j} = 0$ for $j = 1, \dots, n$, with $L = T - V$.

S5. [standard] What is a cyclic (ignorable) coordinate? What is conserved?

Answer in brief: Section 5 — a coordinate that does not appear in the Lagrangian; its conjugate momentum $p_j = \partial L/\partial \dot{q}_j$ is a constant of motion.

S6. [standard] State Hamilton's principle.

Answer in brief: Section 6 — the actual path between two fixed configurations makes the action $S = \int L\,dt$ stationary, $\delta S = 0$.

Long-form questions

L1. [non-AU] State and prove Hamilton's principle. Derive Lagrange's equation from Hamilton's principle. — Christ College, M.Sc Sem-I, CC19PPHY1C01 Classical Mechanics, Nov 2025, Section B Q9.

What to write: Sections 6–7 in full — the action integral and its variation with vanishing end-point variations, integration by parts, and the resulting Lagrange's equations for each generalised coordinate.

L2. [non-AU] Show that the generalised momentum conjugate to a cyclic coordinate is a constant of motion. — St. Aloysius, M.Sc Sem-I, Nov 2021, Part B Q12.

What to write: Section 5 in full — if q_j does not appear in L, then $\partial L/\partial q_j = 0$ and Lagrange's equation gives $d/dt(\partial L/\partial \dot{q}_j) = 0$, so p_j is conserved.

L3. [non-AU] Obtain the Lagrange's equation of motion in terms of Rayleigh (dissipation) function. — St. Aloysius Nov 2021, Part B Q11.

What to write: Section 4 in full — the Rayleigh function $\mathcal{F} = \frac{1}{2}\sum k_{jk}\dot{q}_j\dot{q}_k$ and the equation $\frac{d}{dt}(\partial L/\partial \dot{q}_j) - \partial L/\partial q_j + \partial \mathcal{F}/\partial \dot{q}_j = 0$, with a damped-oscillator example.

L4. [non-AU] Set up the Lagrangian and obtain Lagrange's equations for a simple pendulum and deduce the formula for its time period; also obtain the Lagrangian and Lagrange's equations for a charged particle in an electromagnetic field. — Christ College Nov 2025 Section C Q14; St. Aloysius Nov 2021 Part C Q19B / Midnapore College M.Sc Sem-I PHS 101.2 Q5b. (The Hamiltonian of the charged particle is derived in Unit 3, not here.)

What to write: Sections 3–4 in full — pendulum: $L = \frac{1}{2}ml^2\dot{\theta}^2 + mgl\cos\theta$, giving $\ddot{\theta} + (g/l)\sin\theta = 0$ and $T = 2\pi\sqrt{l/g}$; charged particle: $L = \frac{1}{2}mv^2 - q\phi + q\mathbf{v}\cdot\mathbf{A}$, giving the Lorentz force $m\dot{\mathbf{v}} = q(\mathbf{E} + \mathbf{v}\times\mathbf{B})$.

Sources

These notes were written from the unit syllabus only, as the user asked. No external PDF, textbook, or lecture notes were used. The questions are from the Assam University paper bank and other universities' papers (see the Questions section above), not from any source used for the notes.